

$$\begin{array}{c}
\mathcal{K}_{0^+}^{\#1} \quad \mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{0^+}^{\#1} \dagger \quad \begin{array}{|c|c|} \hline -\frac{\Upsilon_1 k^2}{2} & 0 \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#1} \dagger^{\alpha\beta\chi} \quad \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \quad \mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta} \quad \mathcal{K}_{1^+}^{\#1}{}_{\alpha} \quad \mathcal{K}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta} \quad \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta} \quad \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha} \quad \begin{array}{|c|c|c|c|} \hline 0 & 0 & -\frac{k^2}{6} \frac{\kappa_{15}^{(4)}}{\sqrt{2}} & -\frac{k^2}{3} \frac{\kappa_{15}^{(4)}}{\sqrt{2}} \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha} \quad \begin{array}{|c|c|c|c|} \hline 0 & 0 & -\frac{k^2}{3} \frac{\kappa_{15}^{(4)}}{\sqrt{2}} & -\frac{k^2}{3} \frac{\kappa_{15}^{(4)}}{\sqrt{2}} \\ \hline \end{array} \quad \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta} \quad \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi} \quad \begin{array}{|c|c|} \hline 0 & \frac{3k^2}{2} \frac{\kappa_{15}^{(4)}}{\sqrt{2}} \\ \hline \end{array}
\end{array}$$

$$\begin{array}{c}
\mathcal{J}_{0^+}^{\#1} \quad \mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{0^+}^{\#1} \dagger \quad \begin{array}{|c|c|} \hline \frac{1}{\text{Det}(0^+)} & 0 \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#1} \dagger^{\alpha\beta\chi} \quad \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \quad \mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta} \quad \mathcal{J}_{1^+}^{\#1}{}_{\alpha} \quad \mathcal{J}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta} \quad \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta} \quad \begin{array}{|c|c|c|c|} \hline 0 & 0 & 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha} \quad \begin{array}{|c|c|c|c|} \hline 0 & 0 & -\frac{2}{3k^2} \frac{\kappa_{15}^{(4)}}{\sqrt{2}} & -\frac{2}{3k^2} \frac{\kappa_{15}^{(4)}}{\sqrt{2}} \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha} \quad \begin{array}{|c|c|c|c|} \hline 0 & 0 & -\frac{2}{3k^2} \frac{\kappa_{15}^{(4)}}{\sqrt{2}} & -\frac{4}{3k^2} \frac{\kappa_{15}^{(4)}}{\sqrt{2}} \\ \hline \end{array} \quad \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta} \quad \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta} \quad \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta\chi} \quad \begin{array}{|c|c|} \hline 0 & \frac{2}{3k^2} \frac{\kappa_{15}^{(4)}}{\sqrt{2}} \\ \hline \end{array}
\end{array}$$

Abbreviations used in matrices

$$\Upsilon_1 = 3 \kappa_{15}^{(4)} + 5 \kappa_{15}^{(4)} \quad \&\& \text{Det}(0^+) = -\frac{1}{2} k^2 (3 \kappa_{15}^{(4)} + 5 \kappa_{15}^{(4)})$$

Lagrangian

$$\begin{aligned}
& \kappa_{15}^{(4)} \partial_{\beta} \mathcal{K}_{\chi\delta}^{\delta} \partial^{\chi} \mathcal{K}_{\alpha}^{\alpha\beta} + \frac{5}{3} \kappa_{15}^{(4)} \partial_{\chi} \mathcal{K}_{\beta\delta}^{\delta} \partial^{\chi} \mathcal{K}_{\alpha}^{\alpha\beta} + 2 \kappa_{15}^{(4)} \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}_{\alpha\chi}^{\delta} + \\
& 2 \kappa_{15}^{(4)} \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \kappa_{15}^{(4)} \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}_{\chi\alpha}^{\delta} - 4 \kappa_{15}^{(4)} \partial^{\chi} \mathcal{K}_{\alpha}^{\alpha\beta} \partial_{\delta} \mathcal{K}_{\chi\beta}^{\delta} - \\
& \kappa_{15}^{(4)} \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}_{\beta\chi}^{\delta} + \kappa_{15}^{(4)} \partial_{\delta} \mathcal{K}_{\alpha\beta\chi} \partial^{\delta} \mathcal{K}^{\alpha\beta\chi} + \kappa_{15}^{(4)} \partial_{\delta} \mathcal{K}_{\beta\alpha\chi} \partial^{\delta} \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{0^+}^{\#1\alpha\beta\chi} = 0$	1	$ \begin{aligned} & \partial_{\delta} \partial^{\alpha} \mathcal{J}^{\beta\chi\delta} + \partial_{\delta} \partial^{\alpha} \mathcal{J}^{\delta\beta\chi} + \partial_{\delta} \partial^{\beta} \mathcal{J}^{\chi\alpha\delta} + \partial_{\delta} \partial^{\chi} \mathcal{J}^{\alpha\beta\delta} + \partial_{\delta} \partial^{\chi} \mathcal{J}^{\delta\alpha\beta} + \partial_{\delta} \partial^{\delta} \mathcal{J}^{\beta\alpha\chi} = \\ & \partial_{\delta} \partial^{\alpha} \mathcal{J}^{\chi\beta\delta} + \partial_{\delta} \partial^{\beta} \mathcal{J}^{\alpha\chi\delta} + \partial_{\delta} \partial^{\beta} \mathcal{J}^{\delta\alpha\chi} + \partial_{\delta} \partial^{\chi} \mathcal{J}^{\beta\alpha\delta} + \partial_{\delta} \partial^{\delta} \mathcal{J}^{\alpha\beta\chi} + \partial_{\delta} \partial^{\delta} \mathcal{J}^{\chi\alpha\beta} \end{aligned} $
$\mathcal{J}_{1^+}^{\#1\alpha} = \mathcal{J}_{1^+}^{\#2\alpha}$	3	$\partial_{\chi} \partial^{\alpha} \mathcal{J}_{\beta}^{\beta\chi} + \partial_{\chi} \partial^{\chi} \mathcal{J}_{\beta}^{\beta\alpha} = 0$
$\mathcal{J}_{1^+}^{\#2\alpha\beta} = 0$	3	$\partial_{\delta} \partial_{\chi} \partial^{\alpha} \mathcal{J}^{\chi\beta\delta} + \partial_{\delta} \partial^{\delta} \partial_{\chi} \mathcal{J}^{\chi\alpha\beta} = \partial_{\delta} \partial_{\chi} \partial^{\beta} \mathcal{J}^{\chi\alpha\delta}$
$\mathcal{J}_{1^+}^{\#1\alpha\beta} = 0$	3	$\partial_{\delta} \partial_{\chi} \partial^{\alpha} \mathcal{J}^{\chi\beta\delta} + \partial_{\delta} \partial^{\delta} \partial_{\chi} \mathcal{J}^{\beta\alpha\chi} = \partial_{\delta} \partial_{\chi} \partial^{\beta} \mathcal{J}^{\chi\alpha\delta} + \partial_{\delta} \partial^{\delta} \partial_{\chi} \mathcal{J}^{\alpha\beta\chi}$
$\mathcal{J}_{2^+}^{\#1\alpha\beta} = 0$	5	$3 \partial_{\delta} \partial_{\chi} \partial^{\alpha} \mathcal{J}^{\chi\beta\delta} + 3 \partial_{\delta} \partial_{\chi} \partial^{\beta} \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \mathcal{J}_{\chi}^{\chi\delta} = 2 \partial_{\delta} \partial^{\beta} \partial^{\alpha} \mathcal{J}_{\chi}^{\chi\delta} + 3 (\partial_{\delta} \partial^{\delta} \partial_{\chi} \mathcal{J}^{\alpha\beta\chi} + \partial_{\delta} \partial^{\delta} \partial_{\chi} \mathcal{J}^{\beta\alpha\chi})$
Total # constraint(s):	15	
Resolved unitarity condition(s):	True	